

Beyond Transactions: Sustaining Rural Mutual Aid through Mediated Reciprocity

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Abstract

Rural China faces rapid aging as out-migration weakens family support and formal services remain inaccessible. Community-based mutual aid emerges as a critical alternative, yet transaction-oriented models consistently fail. Through fieldwork in two contrasting villages involving 38 interviews and a participatory workshop, we investigate why quantifying care erodes the relational logic of rural mutual aid. We identify three dilemmas: weakening connections where depopulation makes needs invisible, suppressed help-seeking driven by dignity concerns and fear of unrepayable debt, and erosion of motivation as helping behaviors become socially invisible. Findings reveal that transaction-oriented platforms commodify care, creating “cold” exchanges that clash with “warm” relational norms. We propose a mediated reciprocity framework with three concepts: Relational Routing, Voice Mediation, and Symbolic Recognition Loops. This work contributes empirical insights into the crowding-out effect and proposes design principles that preserve relational warmth while enabling mutual aid.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Collaborative and social computing*.

Keywords

Rural aging, Mutual-aid eldercare, Time banking, Crowding-out effect, Social recognition, Dignity, Community platforms, China

ACM Reference Format:

Yunqi Guo, Lan Zeng, Lixing He, Yingfang Wang, Zhuohan Huang, Ruihan Xie, Chunhao Huang, and Guoliang Xing. 2026. Beyond Transactions: Sustaining Rural Mutual Aid through Mediated Reciprocity. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799061>

1 Introduction

Rural China is experiencing rapid population aging as sustained out-migration leaves many older adults “left behind.” In 2020, adults aged 60+ accounted for 23.8% of the rural population—nearly eight percentage points higher than urban areas [13]. As traditional family support weakens and formal services remain prohibitively expensive or inaccessible, *community-based mutual-aid eldercare*—neighbor-to-neighbor support coordinated via local social networks—has emerged as a critical alternative.

However, existing mutual-aid approaches remain poorly aligned with rural conditions. Transaction-oriented models (Time Banks [12], monetized platforms) attempt to quantify care, transforming “warm” relational exchanges into “cold” transactional ones and eroding the relational logic that historically sustained rural cooperation. Unstructured social media channels, such as WeChat groups [6], suffer from coordination overload. Prior HCI work [14] focuses on urban contexts, leaving a gap in understanding how to sustain mutual aid in resource-constrained rural settings.

To investigate these issues and develop solutions rooted in local value systems, we address the following research question: **What barriers prevent sustainable mutual aid in rural eldercare communities, and how can technology support mutual aid without commodifying it?** We examined this question through



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799061>

qualitative fieldwork in two Chinese villages, combining an ethnographic study of 15 households (38 interviewees) with a participatory design workshop. Our investigation reveals three critical structural dilemmas that disrupt the flow of help:

- **Weakening Connections**—Depopulation has sharply reduced incidental encounters that once anchored neighborly support, creating a demand discovery failure where needs become invisible to would-be helpers.
- **Suppressed Help-Seeking**—Elders refrain from requesting help due to dignity concerns and fear of imposing unrepayable debt in shrinking reciprocal networks, creating a dignity-preserving mechanism gap.
- **Erosion of Motivation**—Helping behaviors become socially invisible in depopulating villages, creating a recognition gap where the community acknowledgment that historically sustained neighborly support is absent.

These dilemmas reveal a fundamental misalignment: quantifying care destroys the social capital these systems aim to mobilize. This paper contributes empirical evidence of why mutual aid models fail in rural contexts and proposes a “Mediated Reciprocity” framework implemented as a WeChat mini program with three core features: *Relational Routing* (trust-based scoping to Inner Circle), *Voice Mediation* (voice-first, anonymized requests), and *Symbolic Recognition Loops* (public Community Vitality display). These findings challenge prevailing design assumptions and reveal how smartphone apps can mediate mutual aid without commodifying it.

2 Background and Related Work

Rural aging represents a global challenge, particularly in regions experiencing sustained youth out-migration [4]. In rural China, approximately 59% of elderly do not co-reside with adult children [5], with 45% living alone or with only a spouse, while formal institutional care remains prohibitively expensive (averaging USD \$5,000 annually vs. household incomes of \$1,400–\$2,500 [7]).

Sociologist Fei Xiaotong described rural China as a “Familiar-Person Society” where support is embedded in acquaintance networks rather than transactional services [2]. Recent innovations like “Love Supermarkets” [1] provide immediate material feedback (points for rice/oil) combined with high-visibility social recognition (public honor rolls), revealing that abstract future credits fail to motivate whereas tangible, dignity-enhancing rewards succeed. This aligns with the “crowding-out effect” [3]: introducing transactional incentives erodes intrinsic altruistic motivations.

Current platforms fall into three categories: (1) *Time Banks* [12]—systems that record hours of help given and allow participants to redeem credits for future care—which fail in shrinking populations where future value is uncertain and clash with rural Renqing (favors) that rely on ambiguous, indefinite reciprocity rather than immediate accounting [8]; (2) *Monetized platforms* [10], which commodify neighborly care and are perceived by rural elders as opportunistic rather than altruistic [9]; and (3) *Unstructured social media channels* (e.g., WeChat groups [6]), which fail due to coordination overload and requests being buried in fast-moving threads. Prior HCI surveys [14] focus on urban contexts, while smart eldercare technologies often operate within a surveillance paradigm that treats older adults as passive recipients rather than active contributors [11].

3 Method

To understand why existing mutual aid models face challenges in rural eldercare, we conducted a qualitative field study in two contrasting Chinese villages, combining in-home ethnographic interviews with a participatory design workshop.

3.1 Field Sites

We selected two villages within the same prefecture—to control for regional cultural variation—representing divergent economic trajectories: **Village A** (traditional agricultural, population 1,503, per capita income \$1,400, 33% aged 60+) and **Village B** (semi-industrialized, population 1,523, per capita income \$2,500, 63% aged 65+). This contrast allowed us to examine whether economic development alleviates care burdens while maintaining comparable cultural norms.

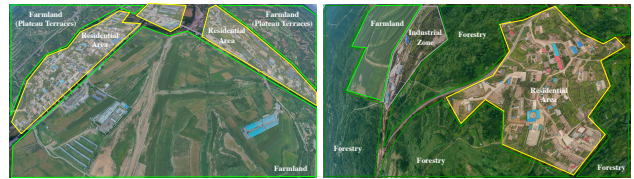


Figure 1: (a) Drone view of Village A: Farmland dominates most of the image, covering both the upper and lower parts, with plateau terraces on the edges. Residential areas are clustered in the central and left portions, occupying a smaller share compared to the farmland. **(b) Drone view of Village B:** The residential area takes up a large portion on the right side, surrounded by forestry that covers most of the remaining space. A small farmland patch and an industrial zone are located in the upper left corner.

3.2 Mixed-Methods Field Study

In-Home Interviews. We conducted semi-structured interviews with 15 households (38 participants: 31 older adults ages 56–94, $M=72.0$; 7 stakeholders including village doctors and cadres). Village cadres function as “human middleware,” bridging digital systems and non-literate elders—revealing that intermediaries are essential in rural contexts. Participants were purposively sampled across living arrangements and functional status (see Figure 2). Interviews (20–60 minutes, 19.5 hours total) addressed our RQ by covering daily care arrangements, experiences with existing mutual aid practices, attitudes toward transaction-based models, barriers to help-seeking and help-giving behaviors, and social connection patterns.

Participatory Workshop. We conducted a half-day participatory workshop in Village A ($N=22$) exploring stakeholders’ perspectives on technology-supported mutual aid. Through scenario discussions, affinity diagramming, and preference voting, we examined reasons for the failure of existing coordination mechanisms, cultural resistance to monetized care (“crowding-out effect”), and values and incentives that could sustain mutual aid. Workshop discussions were audio-recorded and combined with observational field notes.

Data Analysis. Qualitative data were analyzed using reflexive thematic analysis: four researchers independently generated initial

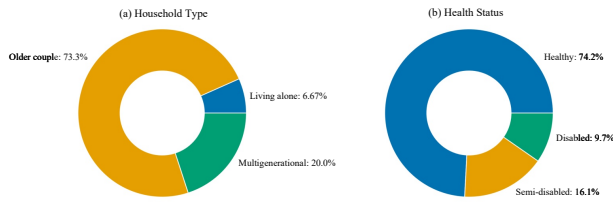


Figure 2: (a) A donut chart showing household types among 31 elders: 73.3% were older couples, 20.0% lived in multigenerational households, and 6.7% lived alone. (b) A donut chart showing health status: 74.2% were healthy, 16.1% were semi-disabled, and 9.7% were disabled (such as due to stroke or requiring a wheelchair).

themes, then met to resolve disagreements and refine codes until consensus was reached.¹

4 Findings and Design Concepts

4.1 Findings: Three Critical Dilemmas in Rural Mutual Aid

Our field study reveals that mutual aid is not failing due to lack of benevolence, but due to three interlinked structural dilemmas that disrupt the flow of help: weakening connections, suppressed help-seeking, and erosion of motivation. These dilemmas all stem from transaction-oriented platforms forcing *market exchange* logic (immediate, equivalent monetary settlement that terminates relationships) into contexts requiring *social exchange* (ambiguous reciprocity that maintains long-term relationships [2]), transforming “warm” relational exchanges into “cold” transactional ones.

Weakening Connections. Depopulation has sharply reduced incidental encounters that once anchored neighborly support. In Village A, gatherings that once hosted 30 neighbors now attract only 5–8, creating a prevailing “quietness.” This disrupts two critical preconditions: (1) *Demand discovery failure*—without the “village gaze” of dense encounters, needs become invisible to willing helpers; (2) *Contextual triggers*—help traditionally emerged from context (e.g., seeing a neighbor struggle), but these visual triggers disappear. As one participant summarized: “We still care about each other, but we simply do not see each other enough.”

Unstructured social media channels, such as WeChat groups, exacerbate this: requests are buried in fast-moving threads or go unnoticed. As a cadre noted, “If I don’t step in to assign it, everyone thinks someone else will do it.” Geographic dispersion creates matching inefficiencies—rural coordination costs are approximately 3.2 times higher than urban contexts [12].

Suppressed Help-Seeking. Even when needs are acute, elders refrain from requesting help due to dignity concerns and fear of creating unrepayable debt. Participants expressed profound reluctance to “trouble others,” fearing that explicit requests would expose frailty or violate reciprocity norms. This reluctance stems from scarce local labor resources—fear of imposing a burden they cannot repay. Additionally, 33.5% report difficulty operating complex digital interfaces [12], compounding help-seeking reluctance.

Time Banks create a *liquidity trap*: high demand but low supply, combined with structural distrust in future redemption. Elders see little value in accumulating credits for a future where “young depositors” are absent. Many initiatives are short-term grants lacking sustainability guarantees—when projects terminate, accumulated credits become worthless. Moreover, credits cannot transfer between villages, creating redemption risk. The accounting logic strips away emotional acknowledgement that fuels reciprocity. As one participant expressed: “If you come for points, don’t come.” This reveals the *crowding-out effect*: extrinsic rewards erode intrinsic motivation. Monetized platforms face stronger resistance—elders felt money “made the relationship feel wrong,” destroying the “warm” relationship.

Erosion of Motivation. Relational thinning and dignity concerns converge to erode motivation. Many elders feel declining social utility—no longer needed by their community. One participant lamented: “When I was young, I helped build the village. Now the village is empty, and it feels like no one needs us anymore.” This loss forms a self-reinforcing cycle: when contributions feel insignificant, social utility erodes, leading to further withdrawal.

Workshop participants emphasized that acknowledgement matters more than material rewards—“being recognized by the community” and “feeling useful” scored higher than “earning money” or “accumulating credits.” Yet existing platforms provide neither immediate recognition nor sustained visibility. Time Bank ledgers operate invisibly, failing to generate public acknowledgement. This creates a double loss: helpers lose social recognition that motivates giving, while seekers lose dignity-preserving context. These findings reveal that HCI must shift from designing *accounting tools* that track debts to designing *acknowledgment stages* that perform gratitude.

4.2 Design Concepts: A Mini Program for Mediated Reciprocity

To address these failures, we propose a WeChat mini program that implements three interaction principles prioritizing dignity, intrinsic motivation, and trusted connections over transactional efficiency. We chose WeChat mini programs because WeChat is already the primary digital platform used by rural elders in China, eliminating the need to download and learn a separate app. Mini programs load instantly within WeChat, reducing technical barriers. Unlike existing Time Bank apps that display credit balances and transaction histories, our design embeds these principles into the app’s core interface. The mini program is organized around a *Home Hub* that centralizes access to three design pillars, each addressing a specific structural dilemma: *Relational Routing* addresses weakening connections, *Voice Mediation* addresses suppressed help-seeking, and *Symbolic Recognition Loops* addresses erosion of motivation (see Figure 3). Together, they form a “Mediated Reciprocity” framework that preserves relational warmth while enabling coordination in depopulating communities.

Relational Routing: Trust-Based Request Scoping. To address the demand discovery failure, the mini program implements *Relational Routing* through a pre-configured “Inner Circle” feature accessible from the Home Hub. During initial setup, elders and their family members identify 5–8 trusted neighbors and kin who

¹The study received IRB exemption as minimal-risk research, and all participants provided informed consent.

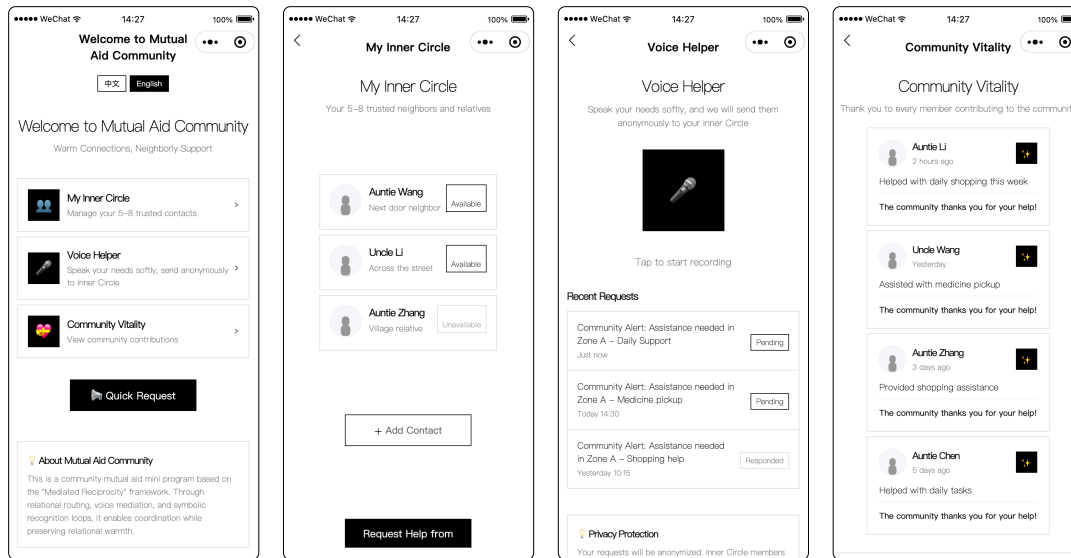


Figure 3: WeChat Mini Program Interface: Interface Design (Left to Right): (1) Home Hub centralizes the three design pillars; (2) Relational Routing scopes alerts to 5–8 trusted neighbors; (3) Voice Mediation converts private audio into standardized system alerts; and (4) Symbolic Recognition Loops visualize non-transactional social praise.

form their “Inner Circle.” When an elder needs help, the app routes requests only to this pre-identified circle, not to the entire village. The interface displays these trusted contacts with familiar names and relationships (e.g., “Auntie Wang—next door neighbor”), mimicking the natural rural “differential mode of association” (Chaxu Geju). This ensures alerts feel personal and actionable, avoiding the “bystander effect” of large chat groups where requests go unnoticed. The app respects existing social networks rather than attempting to create new ones through algorithmic matching, routing requests through pre-existing “warm” relationships to reinforce trust rather than replacing it with “cold” transactional logic.

Voice Mediation: Face-Saving Request Interface. To overcome the fear of losing “face” (Mianzi) via explicit requests, the mini program features a voice-first interface accessible from the Home Hub. Elders activate a “Voice Helper” button that listens for casual, private voice pleas (e.g., “I am out of medicine” or “I need someone to help me get groceries”). The system acts as a neutral third party, converting these private voice inputs into standardized, anonymized notifications (e.g., “Community Alert: Assistance needed in Zone A—Medicine pickup”). The elder’s personal identity is detached from the request, transforming a high-vulnerability “begging” act into a low-vulnerability community alert. Unlike traditional app interfaces that require typing explicit requests or selecting from menus—which feels transactional and demanding—this voice-first approach allows requests to be made casually in privacy, preserving dignity by separating the vulnerable individual from the public request. The interface shows only the anonymized alert to the Inner Circle, not the elder’s name or specific need details.

Symbolic Recognition Loops: Public Contribution Display. To avoid the “crowding-out effect” on intrinsic motivation, the mini program replaces private credit ledgers with a public “Community Vitality” display accessible from the Home Hub. Instead of showing

individual credit balances (which elders found demotivating), the app features a village-level screen showing recent contributions in a non-cumulative, rotating format. When someone helps, their contribution appears with symbolic recognition (e.g., “Auntie Li helped with groceries this week—Thank you!”). The interface emphasizes visibility (public displays, not private ledgers), immediacy (contributions appear immediately), and symbolism (public praise, not monetary credits). Elders can redeem symbolic rewards (household staples like rice or oil) through a simple interface, but these rewards are framed as “community gratitude” rather than “payment for services.” As one participant stated: “Money can buy service, but it cannot buy respect. If the village announces who helped at the monthly meeting, everyone knows I am still useful.” The app makes this recognition visible and immediate, restoring the social utility that motivates mutual aid.

Together, these four interface elements form a “Mediated Reciprocity” mini program that *mediates* mutual aid without *commodifying* it. Unlike Time Bank apps that display credit balances and transaction histories, this design embeds dignity-preserving and recognition-enhancing mechanisms directly into the interface, making “warm” relational values visible and actionable.

5 Discussion

Design Implications. Our findings reveal that digital interfaces compound dignity barriers. While *Voice Mediation* reduces digital literacy requirements, our smartphone-based implementation may still challenge some elders. Future designs could explore dedicated voice-activated devices that eliminate smartphone navigation entirely. Village cadres function as “human middleware,” bridging digital systems and non-literate elders. This intermediary role is often viewed as a “bug,” but we argue it is a *feature*—adding institutional legitimacy that algorithms lack. Future systems should

design for human intermediaries, creating interfaces that support verification work while maintaining transparency.

Limitations and Future Work. This study focuses on two villages in one Chinese region; generalizability requires further investigation. Future work should examine similar dignity-motivation tensions in other depopulating rural communities globally and validate design concepts through prototype deployment.

6 Conclusion

This paper investigates barriers to sustainable mutual aid in rural eldercare communities through fieldwork in two contrasting villages, identifying three critical dilemmas: weakening connections, suppressed help-seeking, and erosion of motivation. Our findings reveal why transaction-oriented platforms fail: they commodify care relationships, transforming “warm” relational exchanges into “cold” transactional ones through the “crowding-out effect” and “liquidity trap.” We contribute empirical evidence that mutual aid failure stems from misalignment between “cold” transactional platform logic and the “warm” relational ethic of rural society, and propose a “Mediated Reciprocity” framework implemented as a WeChat mini program with three core features: Relational Routing, Voice Mediation, and Symbolic Recognition Loops. This demonstrates how smartphone apps can mediate mutual aid without commodifying it, highlighting the necessity of designing for “warm” relational values rather than “cold” transactional efficiency.

Acknowledgments

We thank Mengdi Li and Shuting Li for their careful proofreading of the manuscript. We are grateful to the two communities for their assistance and support during the field study, with special thanks to the local organizers who facilitated our fieldwork.

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